PROBLEM #003

Mathematics

Geometry — Proof

★★★★★ ADVANCED

SUBJECT	LEVEL	TOPICS
Mathematics — Geometry	**Olympiad / RMO Level**	**Similarity · Cyclic Quads · Angle Chasing**

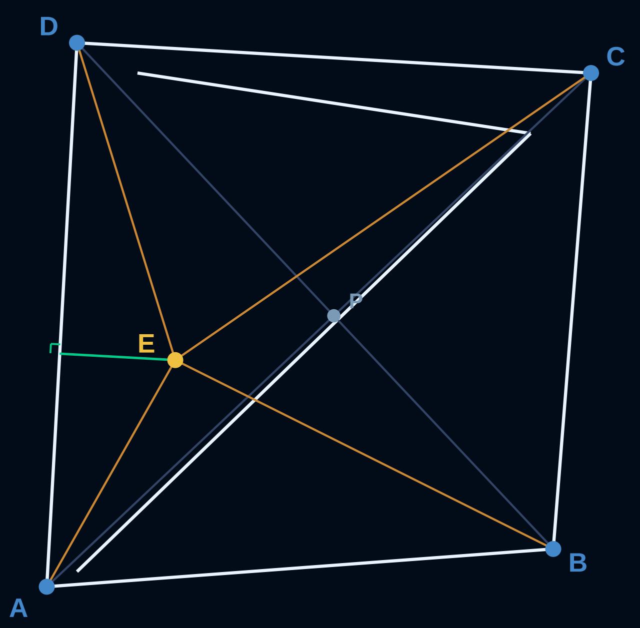
// PAF SCORING

PAF scores will be given based on the solution. Partial credit awarded for significant progress.

THE PROBLEM

Inside convex quadrilateral ABCD, point E was chosen such that
$\angle DAE = \angle CAB$ and $\angle ADE = \angle CDB$.
Prove that if the perpendicular from E to AD passes through the intersection of diagonals of ABCD, then $\angle AEB = \angle CED$.

FIGURE



GIVEN

ABCD is convex quadrilateral
E is inside ABCD
$\angle DAE = \angle CAB$
$\angle ADE = \angle CDB$
Perp from E to AD passes
through diagonals intersection

PROVE

$\angle AEB = \angle CED$

KEY CONCEPTS

- Angle bisector / isogonal
- Triangle similarity (AA)
- Cyclic quadrilaterals
- Angle chasing
- Properties of diagonals

PAF SCORING

Full proof: 100 pts
Key insight: 60 pts
Partial progress: 30 pts
Attempt: 10 pts

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